Unit 25: Sigma Notation and Limits of Finite Sums

1. The idea, in plain words

Last unit, you approximated area using rectangles — and the more rectangles you used, the better the estimate got. This unit gives you two things: a compact way to write "add up a bunch of terms" (sigma notation), and the tool to actually push the number of rectangles to infinity and get an exact answer instead of just an estimate.

Sigma notation is just shorthand for a sum. Instead of writing out $1+2+3+4+5$, you can write $\sum_{i=1}^{5} i$ — read "the sum, as i goes from 1 to 5, of i." The letter i is called the index; it starts at the bottom number, increases by 1 each step, and stops at the top number, plugging into whatever expression sits after the sigma each time.

Why bother with this notation? Because a handful of famous formulas let you evaluate huge sums — even sums of thousands of terms — instantly, without writing out a single term. These come in especially handy for one particular use: writing a Riemann sum with n rectangles using sigma notation, then taking the limit as $n \to \infty$. When you do that, the jagged rectangle approximation smooths out into the exact area under the curve. This is genuinely the formal definition of the definite integral — you're about to compute a few exact areas using nothing but algebra and limits, the hard way, before the next unit hands you a dramatically faster shortcut.

2. Toolbox

Sigma notation:

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + a_3 + \cdots + a_n$$

Properties of sums:

$$\sum_{i=1}^{n} (a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i \qquad \sum_{i=1}^{n} c \cdot a_i = c \sum_{i=1}^{n} a_i \qquad \sum_{i=1}^{n} c = nc \quad \text{(for a constant } c\text{)}$$

The three key summation formulas (memorize these — they do all the heavy lifting):

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$

Building a Riemann sum in sigma notation (right endpoints), for f on $[a,b]$ with n rectangles:

$$\Delta x = \frac{b-a}{n} \qquad x_i = a + i\Delta x \qquad R_n = \sum_{i=1}^{n} f(x_i)\,\Delta x$$

The definite integral, defined as a limit:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$$

The general strategy for computing an exact area this way:

1. Write Δx and x_i in terms of n and i.
2. Build $f(x_i) \cdot \Delta x$ and simplify it into a clean expression involving i, i^2, and/or i^3.
3. Split the sum using the sum properties, and apply the summation formulas to each piece.
4. Simplify the resulting expression (it will be a formula in terms of n alone).
5. Take the limit as $n \to \infty$ — any term with n in the denominator will vanish, leaving the exact area.

3. Common mistakes

- Miscounting the index bounds when expanding a sum. Double check exactly which values of i are included — from the bottom number all the way through the top number, inclusive.
- Using the wrong summation formula — mixing up the formulas for $\sum i$, $\sum i^2$, and $\sum i^3$. Take a moment to check which power of i you actually have before reaching for a formula.
- Losing track of Δx and x_i's dependence on n. Both change as n changes — make sure every n is carried through the algebra correctly before you try to take the limit.
- Algebra slips when simplifying the messy expression in n. These expressions often have several terms with different powers of n in the denominator — take your time expanding and combining them.
- Forgetting that terms like $\dfrac{1}{n}$ or $\dfrac{1}{n^2}$ vanish as $n \to \infty$. This is exactly what makes the final answer collapse down to a clean number — don't accidentally treat n as if it stays finite.

4. Problem Set

Warm-up

Expand and evaluate each sum directly.

1. $\displaystyle\sum_{i=1}^{5} i$
2. $\displaystyle\sum_{i=1}^{4} i^2$
3. $\displaystyle\sum_{i=1}^{4} (2i+1)$
4. $\displaystyle\sum_{i=1}^{3} i^3$
5. $\displaystyle\sum_{i=1}^{6} 3$
6. $\displaystyle\sum_{i=1}^{5} (i-1)$

Standard

Use the summation formulas (rather than expanding term by term) to evaluate each sum.

7. $\displaystyle\sum_{i=1}^{10} i$

8. $\sum_{i=1}^{10} i^2$
9. $\sum_{i=1}^{6} i^3$
10. $\sum_{i=1}^{8} (3i-2)$
11. $\sum_{i=1}^{5} (i^2+2i)$
12. $\sum_{i=1}^{20} 5$

Challenge

Use the limit-of-Riemann-sums definition (with right endpoints) to find each exact area.

13. Find the exact area under $f(x)=x^2$ on $[0,3]$.
14. Find the exact area under $f(x)=x$ on $[0,4]$.
15. Find the exact area under $f(x)=2x+1$ on $[0,2]$.
16. Find the exact area under $f(x)=x^2$ on $[1,2]$.
17. Evaluate: $\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n}\left(\frac{i}{n}\right)^2$

Applied

18. An object's velocity is $v(t)=2t$ ft/s on $[0,5]$. Use the limit-of-Riemann-sums definition to find the exact total distance traveled.
19. Water flows into a tank at a rate of $r(t)=t^2$ gal/min on $[0,4]$. Use the limit definition to find the exact total gallons collected.
20. A company's marginal revenue is $MR(x)=6x$ dollars per unit on $[0,10]$. Use the limit definition to find the exact total revenue.
21. An object's velocity is $v(t)=3t^2$ ft/s on $[0,4]$. Use the limit definition to find the exact total distance traveled.